

System Boot

System Boot

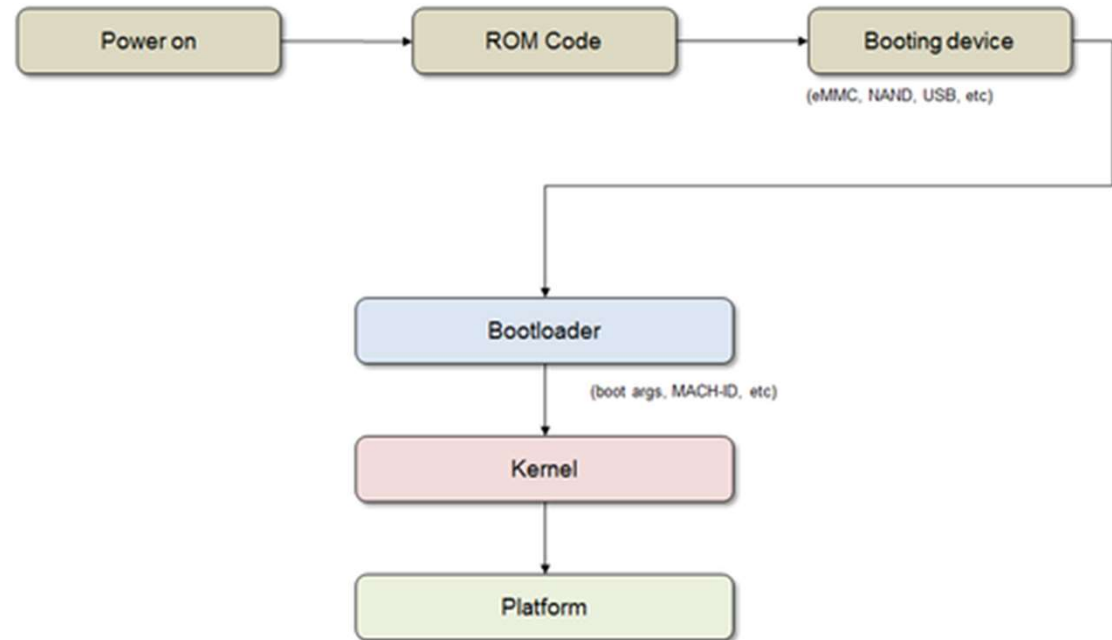
- The boot process is something that
 - happens every time
 - you turn your computer on.
 - You don't really see it,
 - because it happens so fast.
- You press the power button,
 - come back a few minutes later and
 - Windows XP, or Windows 10, or whatever Operating System you use
 - is all loaded.

System Boot

- When power is initialized on system,
 - execution starts at a fixed memory location,
 - **Firmware ROM used to hold initial boot code**
- Operating system must be
 - made available to hardware
 - so hardware can start it

System Boot

- Sometimes two-step process where
 - **Boot block** at fixed location loaded by ROM code,
 - which loads bootstrap program from disk
 - Then loader continues with its job



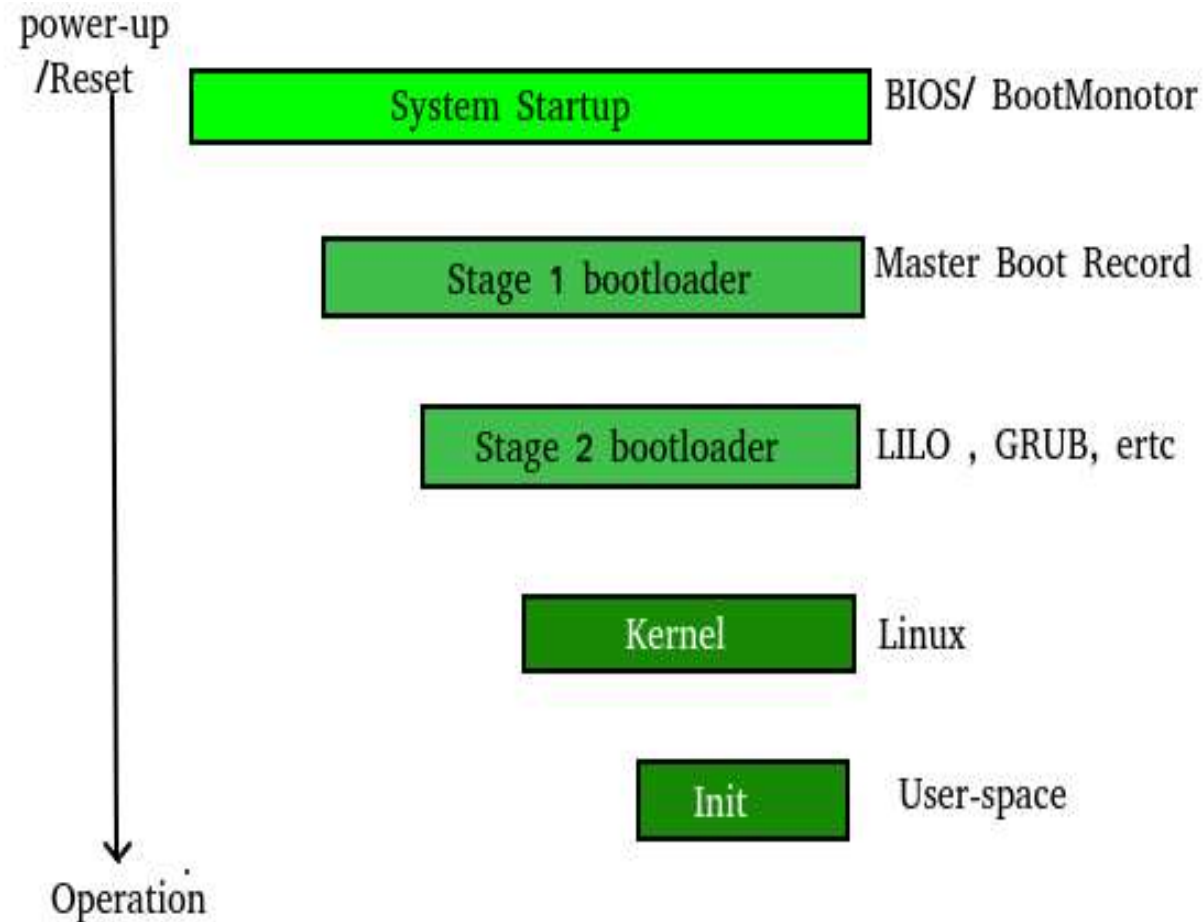
System Boot

- **Bootstrap loader**
 - Small piece of code,
 - stored in **ROM** or **EEPROM** (electrically erasable programmable read-only memory)
 - locates the kernel,
 - loads it into memory, and
 - starts it

System Boot

- Common bootstrap loader,
 - **GRUB**, (the **G**Rand **U**nified **B**ootloader)
 - allows selection of kernel from
 - multiple disks,
 - versions,
 - kernel options
- Kernel loads and
 - system is then **running**

System Boot



System Boot- The working

- The CPU initializes itself
 - after the power in the computer is first turned on.
 - This is done by triggering a series of clock ticks
 - that are generated by the system clock.

System Boot- The working

- After this, the CPU looks for the system's ROM BIOS
 - This first instruction is stored in the ROM BIOS and
 - it instructs the system to
 - run POST (Power On Self Test)
 - in a memory address that is predetermined.
- Since ROM is read only,
 - it cannot be infected by a computer virus.

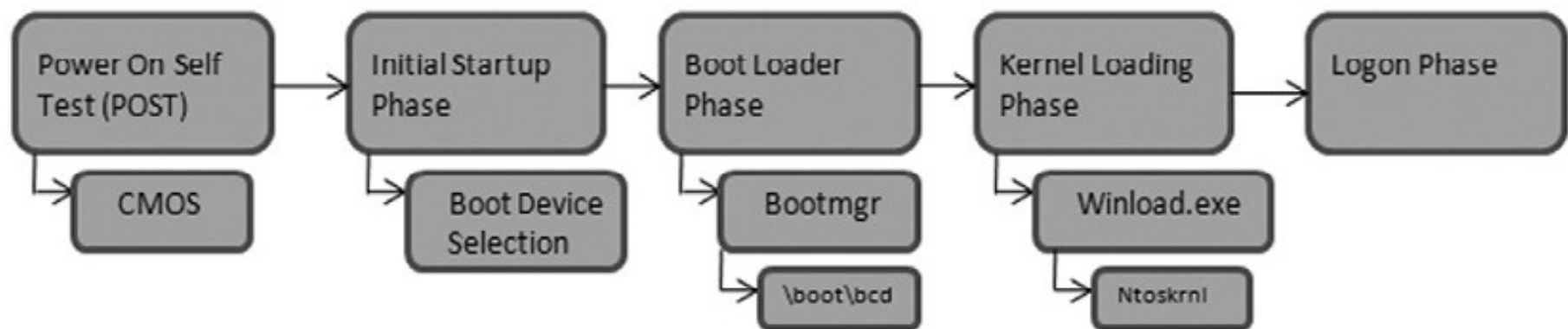
System Boot- The working

- POST first checks
 - the BIOS chip and
 - then the CMOS RAM.
 - If there is no battery failure is detected by POST,
 - then it continues to initialize the CPU.
 - POST also checks
 - the hardware devices,
 - secondary storage devices such as hard drives, ports etc.
 - And other hardware devices such as the mouse and keyboard.
 - This is done to make sure they are working properly.
- 

After POST makes sure that all the components are working properly, then the BIOS start Boot Strap program.

Post method

- To perform this check, the POST
 - Sends out a standard command that says to all the devices, "Check yourselves out!"
 - All the standard devices in the computer then run their own internal diagnostic
 - The POST doesn't specify what they must check.
 - The quality of the diagnostic is up to the people who made that particular device.



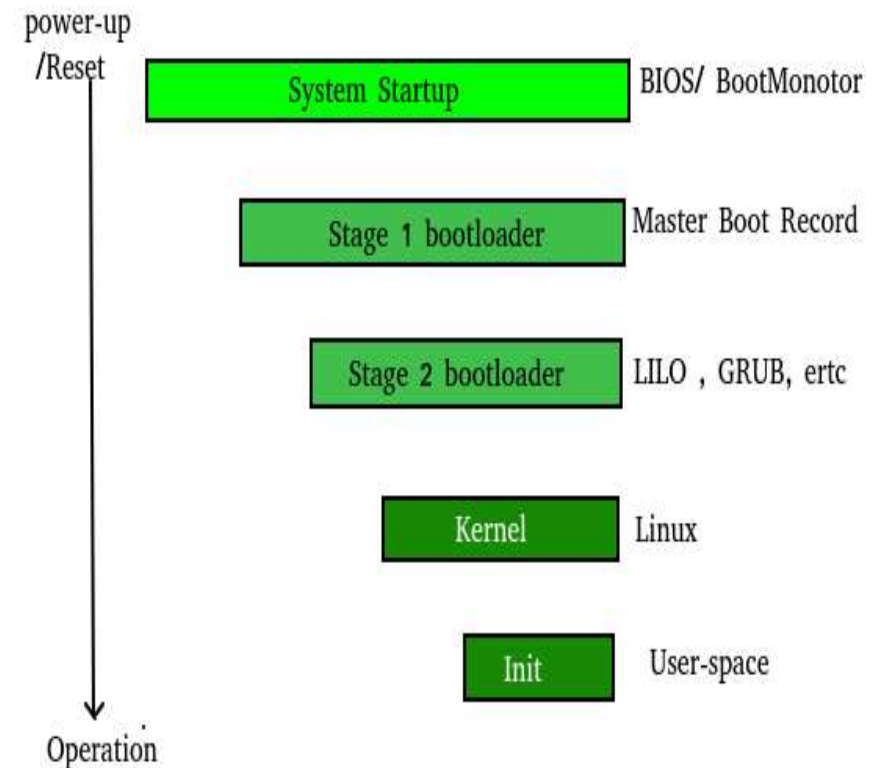
Courtesy : <http://www.c-jump.com/CIS24/Slides/Booting/Booting.html>

System Boot- The working

- For most computers, Boot Strap is stored in BIOS ROM.
 - Changing the Bootstrap code requires changing the ROM hardware chips.

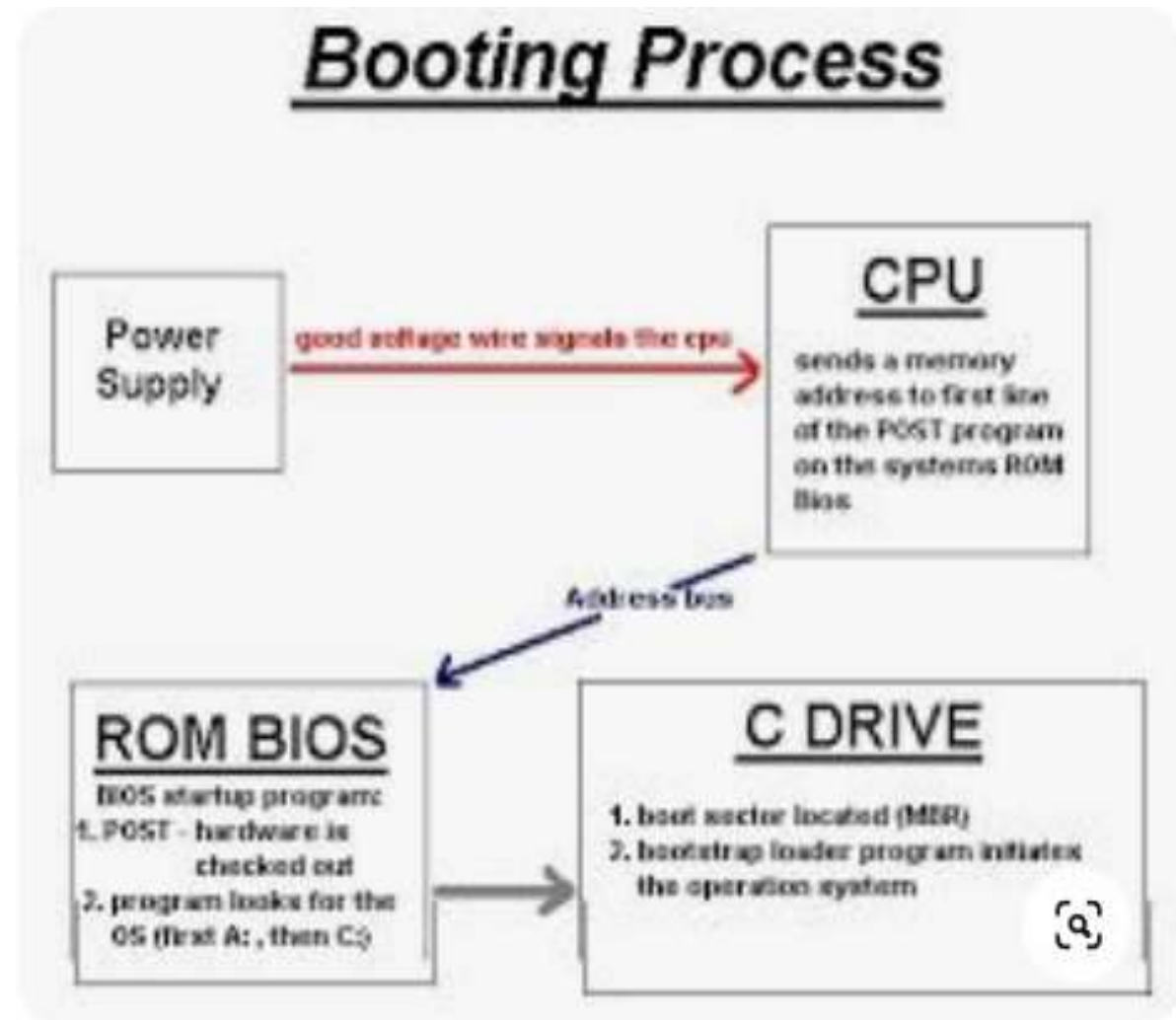
Alternative:

- A tiny Bootstrap loader program is stored in the ROM,
 - whose only job is to bring in a full bootstrap program from the disk.
 - The full bootstrap pgm can be changed easily.



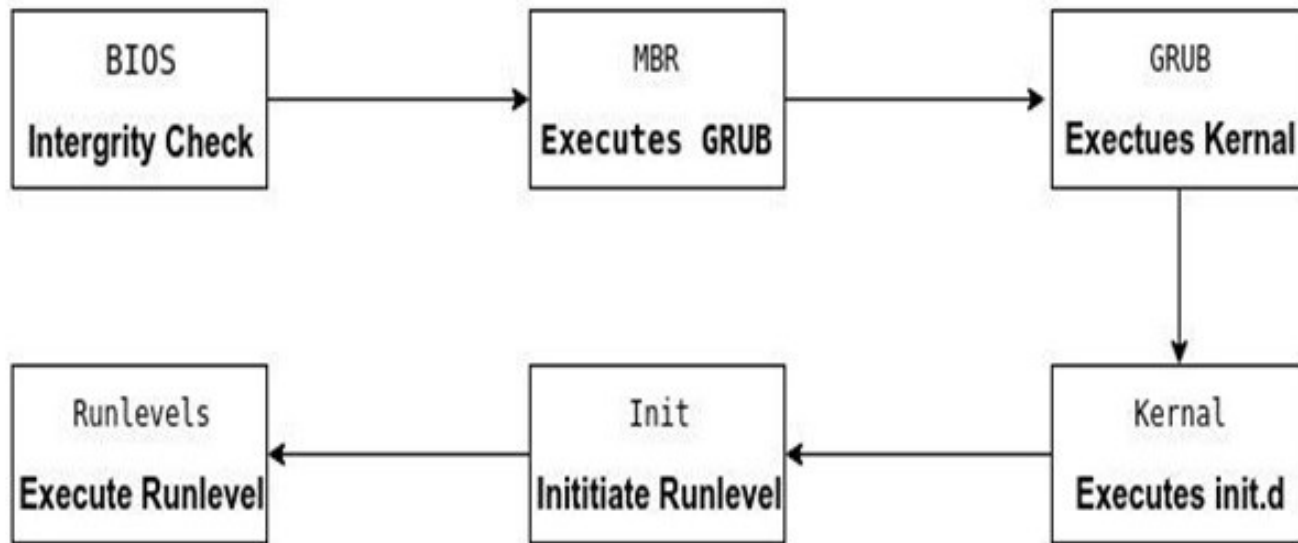
System Boot- The working

- The Bootstrap pgm
 - is stored in a partition called the boot blocks ,
 - at a fixed location on the disk.
- A disk that has a boot partition
 - is called a Boot Disk or System Disk.



System Boot- The working

- The Bootstrap program is loaded into memory and
 - starts its execution.
- The bootstrap program
 - finds the OS kernel on disk,
 - loads that Kernel into memory and
 - jumps to an initial address
 - to begin the OS execution.



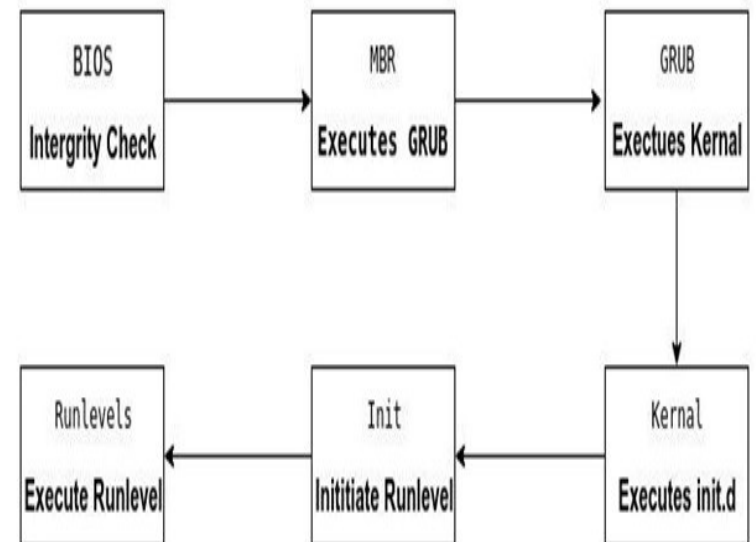
Master Boot Record

The Master Boot Record (MBR) is the information which is in the first sector of any hard disk.

It holds information about GRUB (or LILO in old systems).

init

- This is the last step of the **booting process**.
- It decides the run level by looking at the / etc / inittab file.
- The initial state of the operating system is decided by the run level.



init

Following are the run levels of Operating System:

Level

0 - System Halt

1 - Single user mode

2 - Multiuser, without NFS(Network File System)

3 - full multiuser mode

4 - unused

5 - Full multiuser mode with network and X display manager(X11)

6 - Reboot

We would set the default run level to either 3 or 5.

- The step after it is to
 - 1) Start up various daemons that support networking and other services.
 - 2) X server daemon manages display, keyboard, and mouse.
 - 3) You can see a Graphical Interface and
 - 4) A login screen is displayed during X server daemon is started.

- Daemons are **processes that run unattended**.
- They are constantly in the background and are available at all times.
- Daemons are usually started when the system starts, and they run until the system stops.

Failure during boot

- If the computer cannot boot, we will get a boot failure error.
 - This error indicates that the computer is not passing POST or
 - **a device in the computer, such as the hard drive or memory, has failed.**
 - **You may also hear a beep code to identify which hardware is failing during the POST.**
 - An error message or blue screen may show on the screen as operating system files cannot be loaded, due to not being found or being corrupt.

What is the name given to the process of initializing a microcomputer with its operating system?

- (a) Cold booting
- (b) Booting
- (c) Warm booting
- (d) Boot recording
- (e) None of the above

What is the name given to the process of initializing a microcomputer with its operating system?

- (a) Cold booting
- (b) Booting **(Ans)**
- (c) Warm booting
- (d) Boot recording
- (e) None of the above

- Cold Boot-
 - To perform a cold boot (also called a "hard boot") means to start up a computer that is turned off.
- Warm Boot-
 - It is often used in contrast to a cold boot, which refers to restarting a computer once it has been turned on.
- While a warm boot and cold boot are similar,
 - a cold boot performs a more complete reset of the system than a warm boot.

- Warm Boot-
 - If the computer hangs because of some reason while working, and demands to be restarted to make it working.
 - The process of reset/restart of the computer system is called as warm booting.
 - It is done with the help of reset button or keys (Ctrl+Alt+Del).

Difference between Cold Booting and Warm Booting:

S.NO.	COMPARISON	COLD BOOTING	WARM BOOTING
1.	Initialized by	Power button.	Reset button or by pressing Ctrl+Alt+Del simultaneously.
2.	Performed	Frequent basis.	Not very common.
3.	Alternate names	Hard Booting, Cold start and dead start.	Soft Booting.
4.	POST (Power On Self Test)	Included.	Not included.
5.	Basic	Turning a computer ON from a powerless state.	Resetting a computer from already in running state.
6.	Consequence	Does not affect the data or other hardware.	Can severely affect the system causing the data loss.